

MINGZE PAN

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EDUCATION

Stevens Institute of Technology Systems Engineering Ph.D	<i>Sep 2023 - Present</i> <i>Hoboken, NJ</i>
California State University, Fullerton Computer Engineering M.S.	<i>Aug 2020 - Dec 2022</i> <i>Fullerton, CA</i>
California State University, Fullerton (Exchange Program) Electronic Engineering	<i>Aug 2019 - May 2020</i> <i>Fullerton, CA</i>
Tianjin Normal University Communication Engineering B.S.	<i>Sep 2016 - Jun 2019</i> <i>Tianjin</i>

PUBLICATIONS

Mingze Pan, Sudhanshu Arya, Ying Wang: Domain Knowledge Powered Machine Learning for the Classification of LOS/NLOS Signals for Dedicated-Spectrum SAGIN Networks, DySPAN '24, May , 2024

Mingze Pan, Ying Wang: Bayesian Cooperative LOS/NLOS Classification with Domain Insights and Model Refinement for UAV Communication, IEEE Open Journal of Vehicular Technology, under review

Mingze Pan, Ying Wang: LOS and NLOS Discrimination in SAGIN Networks: A Time-Frequency Representation Approach with Multimodal Integration, IEEE Globecom 2024, under review

Mingshuo Liu, Kevin Han, Shiyi Luo, Mingze Pan, Mousam Hossain, Bo Yuan, Ronald F. DeMara, Yu Bai: An Efficient Video Prediction Recurrent Network using Focal Loss and Decomposed Tensor Train for Imbalance Dataset, GLSVLSI '21, June , 2021

AWARDS

IEEE Student Engineering Team Challenge 2021 Project: Self-Navigating Drone	<i>Second Place</i> <i>Aug 2021</i>
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PROFESSIONAL EXPERIENCE

Stevens Institute of Technology Research Assistant Advisor: Dr. Ying Wang	School of Systems and Enterprises <i>Sep 2023 - Present</i>
• Project1: Developed innovative machine learning models integrating domain knowledge and Bayesian methods for precise LOS/NLOS • Project2: Utilized spectral data analysis for signal characterization, enhancing classification accuracy under diverse environmental conditions • Project3: Implemented a Bayesian network model for real-time detection of vulnerabilities and intrusions within physical communication channels, enhancing network security	
California State University, Fullerton Graduate Student Research Assistant	Computer Engineering Department <i>Jan 2021 - Nov 2021</i>

- **Project1:** Developed a pre-processing module into YOLOv5 to optimize the data set for object detection, which reduced calculation waste
- **Project2:** Implemented edge smart computing with GPS system for drone to autonomously complete extensive analysis of GPS coordinates and adaptive pathfinding

Tianjin Normal University

Communication Engineering Department

National College Innovation and Entrepreneurship Project

Dec 2017 - Jun 2019

Group Leader

Designed an Anti-blocking smoke alarm system for student dormitories in the university.

WORK EXPERIENCE

Stevens Institute of Technology

School of Systems and Enterprises

Research Assistant

Sep 2023 - Present

California State University, Fullerton

Computer Engineering Department

Instructional Student Assistant

Sep 2021 - Nov 2021

TECHNICAL AND LANGUAGE SKILLS

Programming Language

Python, MATLAB, C, C++, Verilog

Software

Visual Studio, HSPICE, Latex, MATLAB, Xilinx Vivado

Operating System

Linux

Data Analysis

Python - Numpy, Pandas, Scikit-Learn, Matplotlib

Deep Learning

Python - Keras, TensorFlow, Pytorch

Language

English, Chinese